

**PYEONGTAEK AB, Korea, Republic of**

WMO: 471270

Lat: 36.967N

Lon: 127.033E

Elev: 16

StdP: 101.13

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 43216

**Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -12.1      | -10.2 | -18.2                         | 0.8 | -8.2 | -16.1 | 0.9 | -6.8 | 9.8                   | -2.7 | 8.6 | -3.0 | 1.8                   | 320  | 0.433 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 8.0                    | 33.1            | 26.1 | 31.9 | 25.3 | 30.6 | 24.7 | 27.6                | 30.4 | 26.7 | 29.7 | 26.0 | 28.9 | 3.3                  | 270  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 27.1                            | 22.8 | 28.8 | 26.1 | 21.5 | 28.1 | 25.2 | 20.3 | 27.4 | 88.4          | 30.7 | 84.4 | 30.1 | 81.1 | 28.1 | 32.0              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 8.2               | 7.1  | 6.1 | -15.4                      | 35.0 | 2.7                | 1.4 | -17.3                                              | 36.0 | -18.9      | 36.9 | -20.4      | 37.6 | -22.3      | 38.7 |    |    |
|                   |      |     | -16.0                      | 29.2 | 2.6                | 1.3 | -17.9                                              | 30.1 | -19.4      | 30.8 | -20.8      | 31.6 | -22.7      | 32.5 |    |    |

**Monthly Climatic Design Conditions**

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 12.7   | -2.4  | 0.2   | 5.6   | 12.2  | 18.1  | 22.6  | 25.9  | 26.4  | 21.4  | 14.6  | 7.1   | -0.1 |
|                                                                               | DBStd       | 10.54  | 4.02  | 3.88  | 3.58  | 3.52  | 2.64  | 2.01  | 2.21  | 2.46  | 2.78  | 3.53  | 4.26  | 4.18 |
|                                                                               | HDD10.0     | 1244   | 384   | 275   | 142   | 17    | 0     | 0     | 0     | 0     | 7     | 107   | 313   |      |
|                                                                               | HDD18.3     | 2803   | 642   | 508   | 394   | 186   | 37    | 1     | 0     | 0     | 7     | 121   | 337   | 571  |
|                                                                               | CDD10.0     | 2234   | 0     | 1     | 7     | 84    | 251   | 378   | 493   | 509   | 341   | 151   | 19    | 0    |
|                                                                               | CDD18.3     | 749    | 0     | 0     | 0     | 2     | 29    | 129   | 235   | 251   | 98    | 6     | 0     | 0    |
|                                                                               | CDH23.3     | 5888   | 0     | 0     | 1     | 29    | 233   | 788   | 1950  | 2288  | 564   | 34    | 1     | 0    |
|                                                                               | CDH26.7     | 1865   | 0     | 0     | 0     | 3     | 34    | 194   | 660   | 854   | 118   | 1     | 0     | 0    |
| Wind                                                                          | WSAvg       | 2.5    | 2.4   | 2.7   | 3.1   | 3.0   | 2.7   | 2.4   | 2.5   | 2.4   | 2.1   | 2.1   | 2.4   | 2.4  |
| Precipitation                                                                 | PrecAvg     | 1275   | 19    | 27    | 46    | 78    | 86    | 142   | 319   | 274   | 162   | 50    | 44    | 29   |
|                                                                               | PrecMax     | 1882   | 57    | 84    | 118   | 184   | 214   | 383   | 697   | 988   | 363   | 202   | 139   | 54   |
|                                                                               | PrecMin     | 756    | 3     | 1     | 10    | 22    | 12    | 23    | 121   | 46    | 16    | 1     | 5     | 8    |
|                                                                               | PrecStd     | 268    | 15    | 20    | 26    | 38    | 44    | 91    | 124   | 197   | 118   | 45    | 30    | 14   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 10.2  | 14.7  | 20.1  | 26.3  | 29.6  | 31.9  | 35.0  | 35.2  | 31.0  | 26.1  | 20.9  | 13.1 |
|                                                                               |             | MCWB   | 6.9   | 11.0  | 12.2  | 16.1  | 18.4  | 21.4  | 26.8  | 26.5  | 23.7  | 18.9  | 15.8  | 9.2  |
|                                                                               | 2%          | DB     | 7.2   | 11.1  | 17.0  | 23.8  | 27.3  | 30.1  | 33.0  | 33.8  | 29.2  | 24.1  | 18.1  | 10.1 |
|                                                                               |             | MCWB   | 4.4   | 6.7   | 10.5  | 15.2  | 17.8  | 21.4  | 26.2  | 26.4  | 22.4  | 17.5  | 13.6  | 7.2  |
|                                                                               | 5%          | DB     | 5.5   | 8.8   | 14.8  | 21.6  | 26.0  | 28.9  | 31.7  | 32.4  | 28.0  | 22.5  | 16.1  | 8.2  |
|                                                                               |             | MCWB   | 3.0   | 5.1   | 9.2   | 13.6  | 17.3  | 21.0  | 25.7  | 25.8  | 21.7  | 16.5  | 12.3  | 5.8  |
|                                                                               | 10%         | DB     | 3.9   | 6.8   | 12.4  | 19.2  | 24.2  | 27.8  | 30.2  | 31.0  | 26.8  | 21.0  | 14.2  | 6.8  |
|                                                                               |             | MCWB   | 1.9   | 3.6   | 7.7   | 12.7  | 16.7  | 20.8  | 25.2  | 25.6  | 20.9  | 15.8  | 10.8  | 4.6  |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 7.6   | 12.0  | 15.0  | 19.7  | 22.5  | 26.0  | 29.0  | 28.9  | 25.9  | 21.7  | 17.3  | 9.9  |
|                                                                               |             | MCDB   | 9.0   | 13.8  | 17.6  | 22.8  | 25.1  | 27.4  | 31.5  | 31.7  | 29.5  | 23.9  | 18.9  | 11.6 |
|                                                                               | 2%          | WB     | 5.1   | 7.3   | 12.2  | 17.2  | 20.6  | 24.4  | 27.7  | 27.8  | 24.4  | 19.5  | 14.8  | 7.9  |
|                                                                               |             | MCDB   | 6.7   | 9.8   | 15.1  | 20.7  | 24.0  | 26.6  | 30.2  | 30.8  | 27.3  | 21.9  | 17.1  | 9.5  |
|                                                                               | 5%          | WB     | 3.5   | 5.4   | 10.1  | 15.5  | 19.3  | 23.2  | 26.8  | 27.0  | 23.4  | 18.0  | 13.0  | 6.2  |
|                                                                               |             | MCDB   | 4.8   | 8.0   | 13.2  | 19.2  | 22.9  | 25.8  | 29.5  | 30.3  | 25.9  | 20.8  | 15.3  | 7.8  |
|                                                                               | 10%         | WB     | 2.0   | 3.9   | 8.3   | 13.8  | 18.1  | 22.2  | 26.1  | 26.2  | 22.4  | 16.8  | 11.5  | 4.7  |
|                                                                               |             | MCDB   | 3.4   | 5.9   | 11.3  | 17.5  | 21.9  | 25.2  | 28.6  | 29.4  | 24.9  | 19.6  | 13.7  | 6.3  |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 9.7   | 10.4  | 11.2  | 12.3  | 11.3  | 9.7   | 7.7   | 8.0   | 9.6   | 11.4  | 10.5  | 9.6  |
|                                                                               |             | MCDBR  | 10.8  | 13.2  | 14.9  | 15.8  | 13.9  | 11.9  | 9.7   | 9.6   | 10.7  | 12.4  | 12.3  | 11.3 |
|                                                                               |             | MCWBR  | 8.3   | 9.3   | 9.0   | 8.2   | 6.3   | 5.1   | 4.5   | 4.2   | 5.2   | 6.7   | 8.3   | 8.8  |
|                                                                               | 5% WB       | MCDBR  | 9.7   | 11.6  | 12.5  | 12.5  | 10.9  | 9.0   | 8.0   | 8.3   | 8.7   | 10.2  | 10.7  | 10.6 |
|                                                                               |             | MCWBR  | 8.2   | 9.0   | 8.8   | 8.0   | 6.5   | 5.2   | 4.6   | 4.4   | 5.3   | 6.6   | 8.2   | 8.8  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.442  | 0.490 | 0.572 | 0.607 | 0.622 | 0.693 | 0.652 | 0.579 | 0.507 | 0.494 | 0.478 | 0.439 |      |
|                                                                               | taud        | 1.940  | 1.858 | 1.711 | 1.683 | 1.702 | 1.623 | 1.758 | 1.928 | 2.054 | 1.961 | 1.946 | 1.964 |      |
|                                                                               | Ebn at Noon | 695    | 715   | 695   | 702   | 702   | 652   | 676   | 717   | 748   | 704   | 653   | 664   |      |
|                                                                               | Edh at Noon | 141    | 174   | 220   | 239   | 238   | 257   | 224   | 185   | 155   | 154   | 140   | 130   |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 2.47   | 3.25  | 4.29  | 4.99  | 5.59  | 5.17  | 4.17  | 4.39  | 4.11  | 3.60  | 2.47  | 2.17  |      |
|                                                                               | RadStd      | 0.15   | 0.27  | 0.43  | 0.38  | 0.52  | 0.50  | 0.61  | 0.63  | 0.45  | 0.27  | 0.26  | 0.15  |      |

**Historical Trends**

|      |                         | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                         | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)     | (m)     |         |
| (46) | Station Only            | N/A   | N/A     | -1.25  | N/A   | N/A     | N/A   | N/A | N/A         | +111    | +66     |         |
| (47) | Regional (49 neighbors) | N/A   | N/A     | -0.81  | N/A   | N/A     | N/A   | N/A | N/A         | +92     | +51     |         |

Nomenclature: See separate page